



Next generation of Microinverter Solutions

Infineon Cycloconverter Board

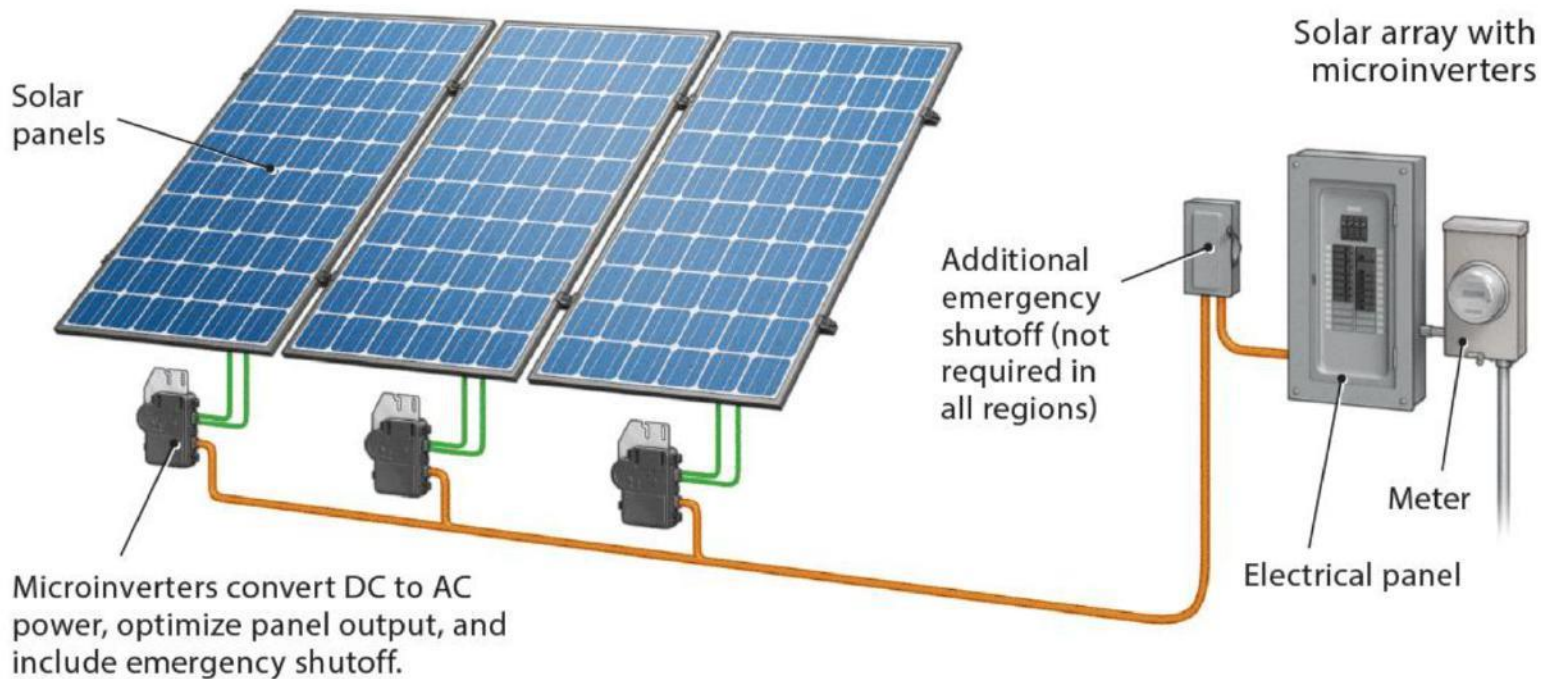
2025.02.03



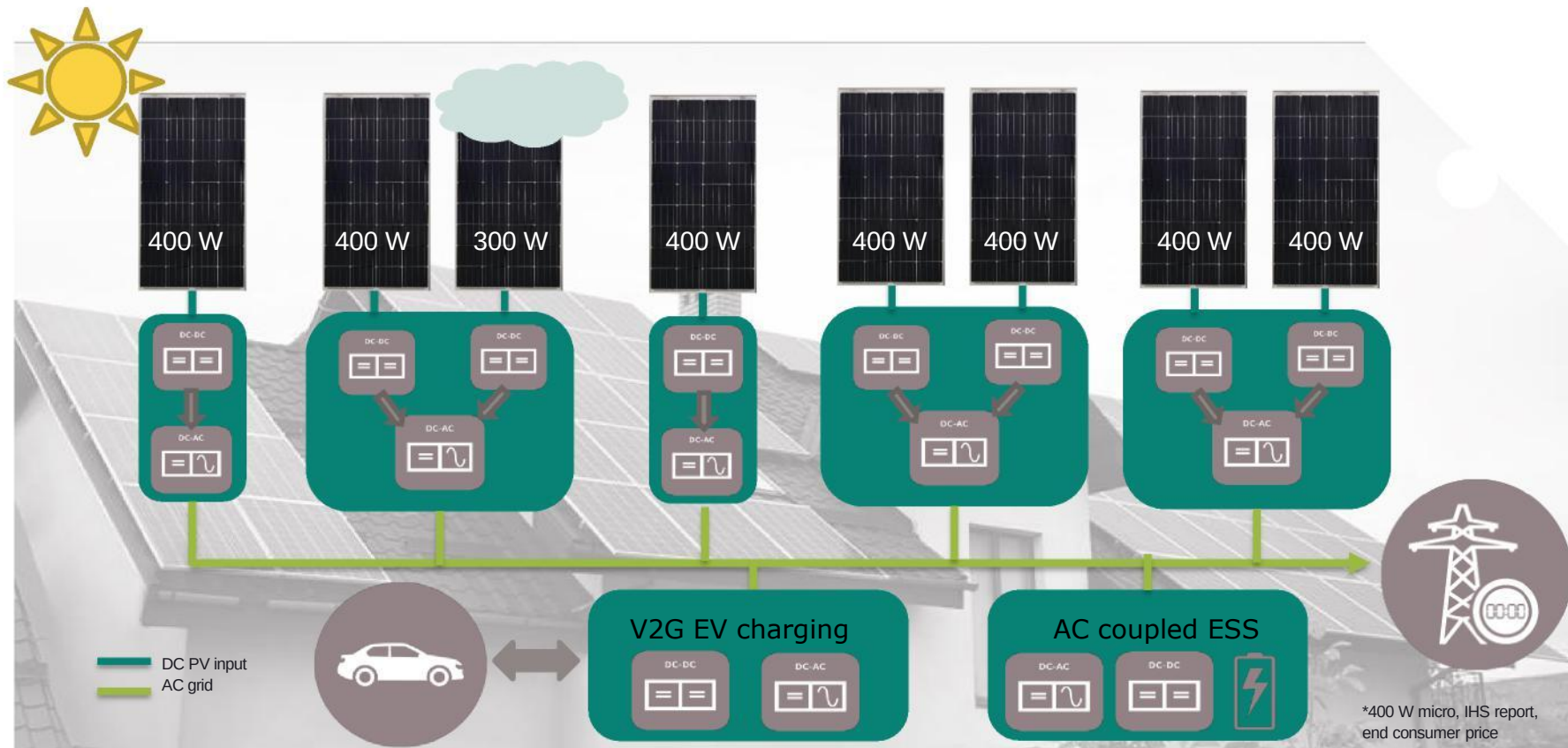
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- 2 Cycloconverter operation and control
- 3 Board and experimental measurements
- 4 Comparison with previous 300W power station

What is a microinverter?



Use case 1: Micro inverter + AC coupled ESS & V2G EV charging



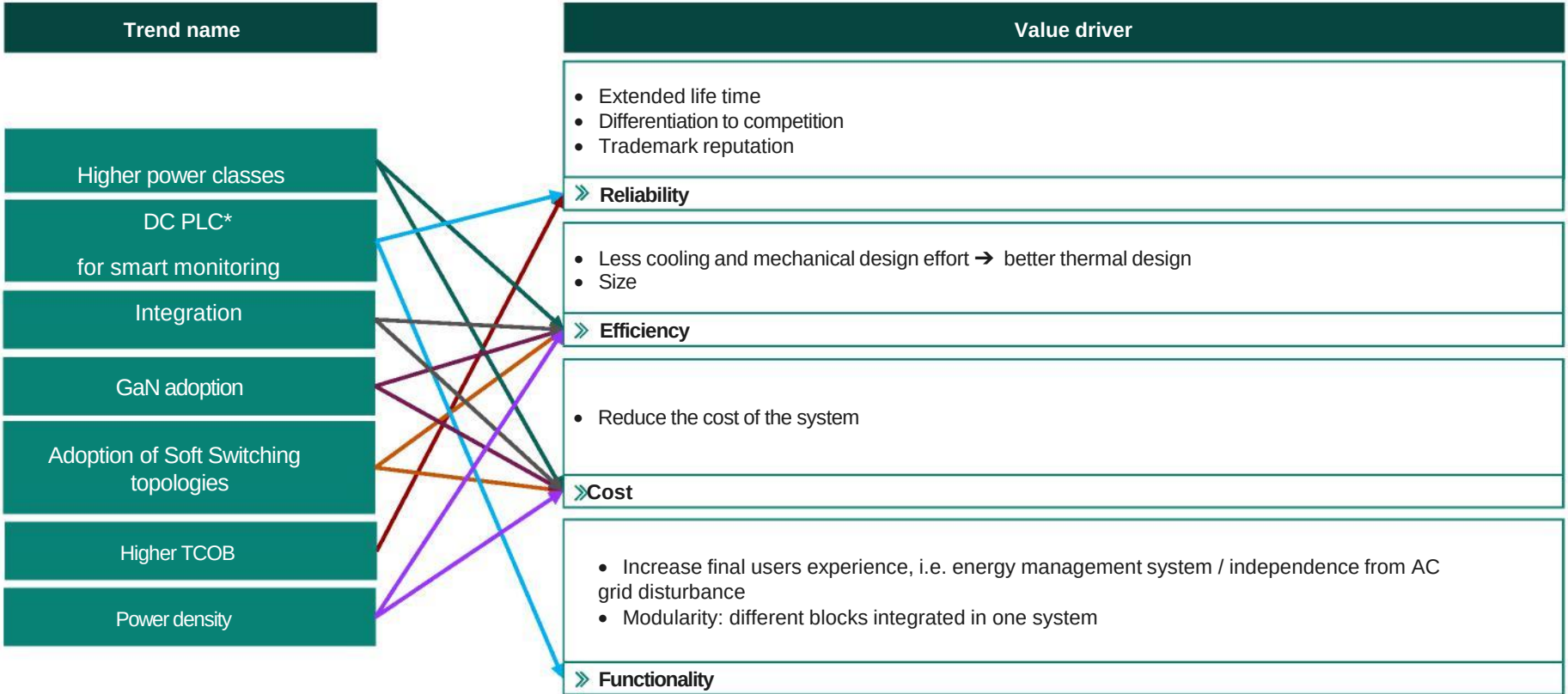
Use case 2: Balcony PV system



— Up to 800W

*1kWh portable ESS
**800 W microinverter

General technical trends in microinverter systems (excerpt from Application Roadmap L1 Res. Solar & ESS)



Microinverter requirements

Microinverter requires 25 years life-time

Grid Synchronization, compatibility with local power grid

Maximum Power Point Tracking

Enable high efficiency (>97%) to reduce heat dissipation & BoM cost

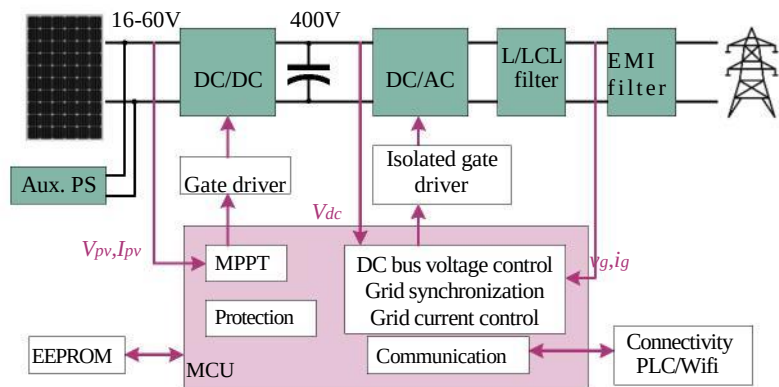
TCoB > 10000 cycles

Harsh environments, T_{amb} up to 60° C – 70° C

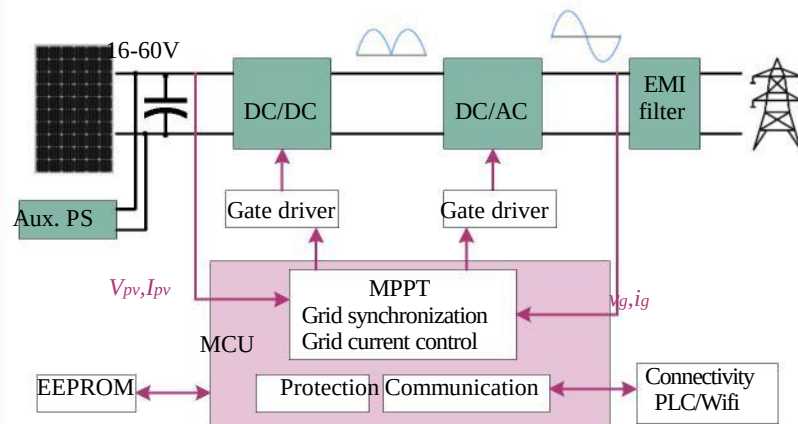
Certification: CA Rule 21 (UL 1741-SA), UL 62109-1, UL1741/IEEE1547, FCC Part 15 Class B, ICES-0003 Class B, CAN/CSA-C22.2 NO. 107.1-01, NEC 2020 and C22.1-2018 Rapid Shutdown Rule

Microinverter architectures (2-stages)

Voltage Source Inverter



Current Source Inverter

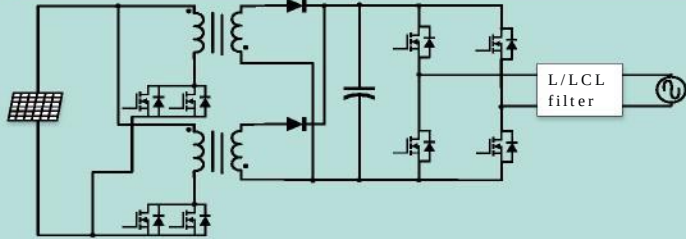


Microinverter topologies overview:

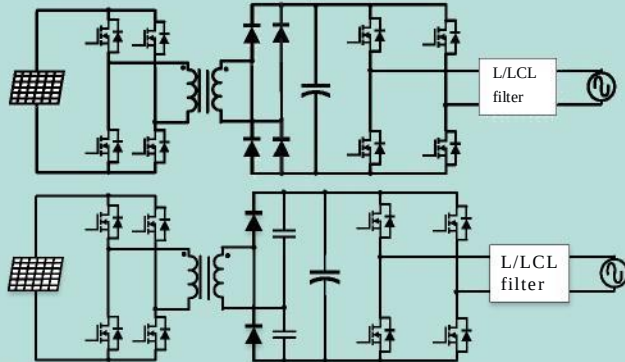
Voltage-source inverters and current-source inverters

Voltage-source inverters

Flyback + Full-Bridge

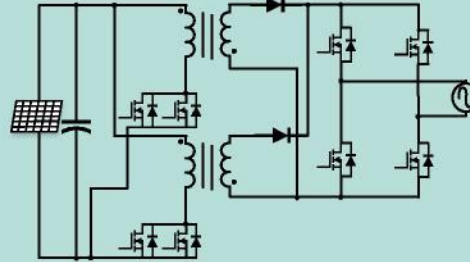


Phase-Shift Full Bridge + Full-Bridge

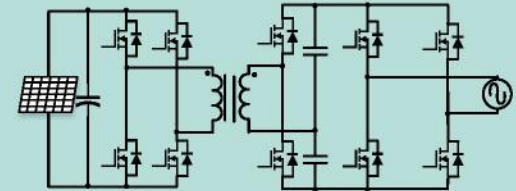


Current-source inverters

Flyback + Unfolder



DAB + Unfolder



Cycloconverter

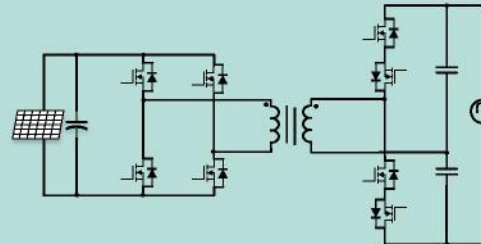


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- 3 Board and experimental measurements
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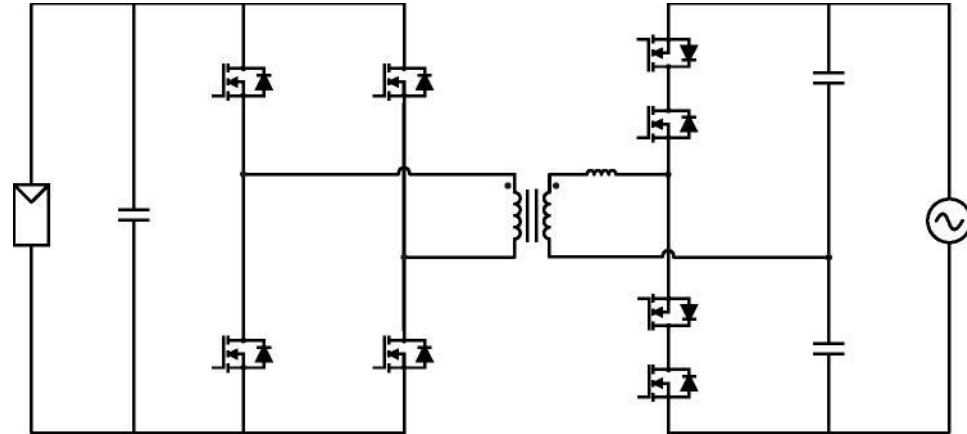
Microinverter topologies overview:

Single-stage cycloconverter

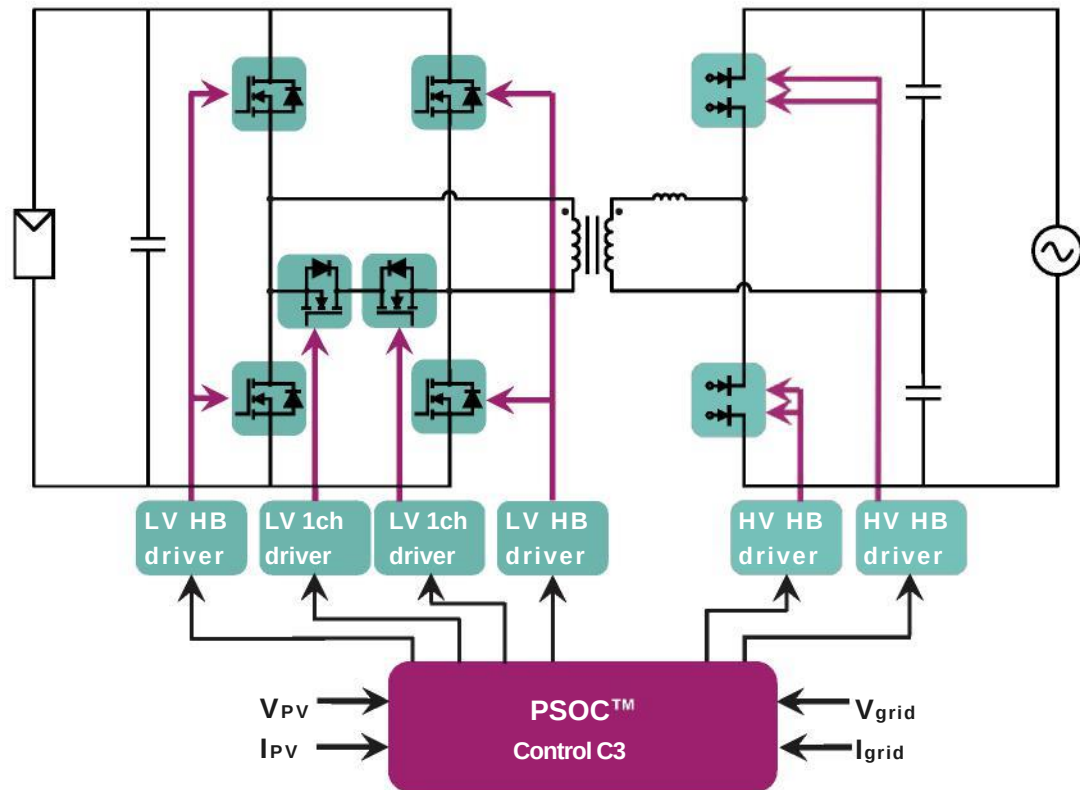
- Single-stage microinverter based on cycloconverter topology

- Compared to two-stage topologies cycloconverter offers several advantages:
 - Higher efficiency up to 2%
 - Higher power density
 - Reactive power capability
 - Simpler/smaller transformer (resonant type, which is smaller compared to flyback type)
 - Fewer semiconductor components required

Classic Cycloconverter Topology



Infineon products in cycloconverter system

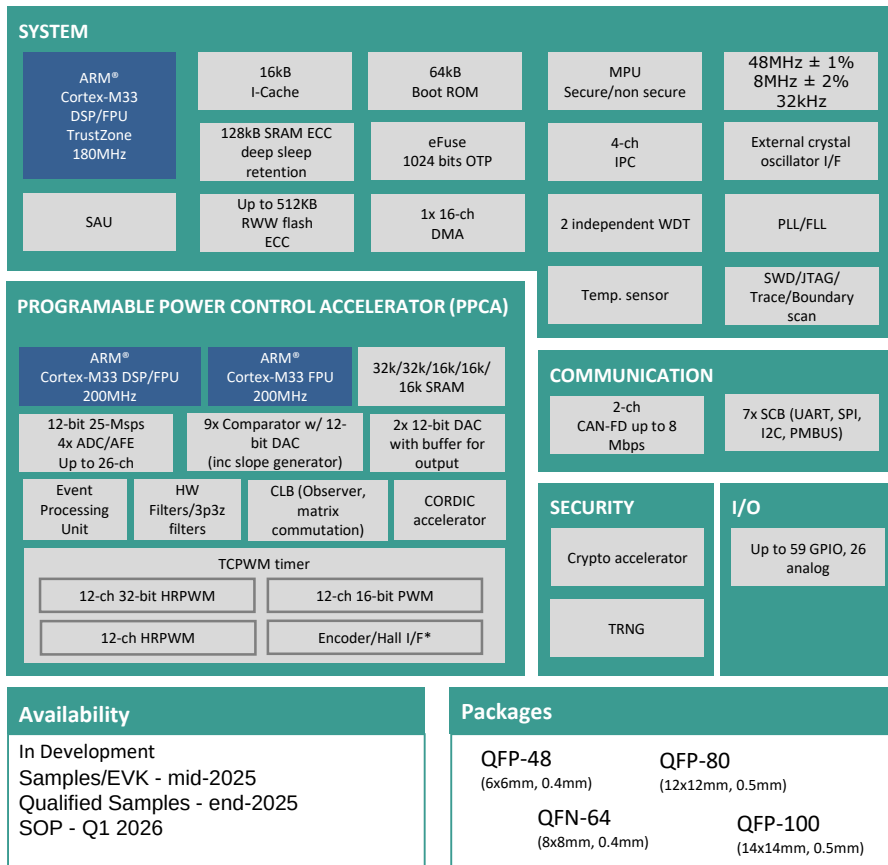


Infineon components for cycloconverter solution

LV MOSFET	4x ISC018N08NM6 (FB) 2x ISC056N08NM6 (B2B)
HV GaN BDS	IGLT65R055B2
LV GD	2EDL8034F5B (Si HB) 1EDN8550B (Si 1-ch)
HV GD	2EDR7259X (GaN BDS)
MCU	PSOC™ Control C3 Main Line: PSC3M5FDS2AFQ1

PSOC™ Control C3 Performance Line

180MHz Single Core with dual 180MHz PPCA



Description

Applications / Target Markets

- Server & Telcom SMPS, Solar Inverters, LV Charger, AC/DC, DC/DC

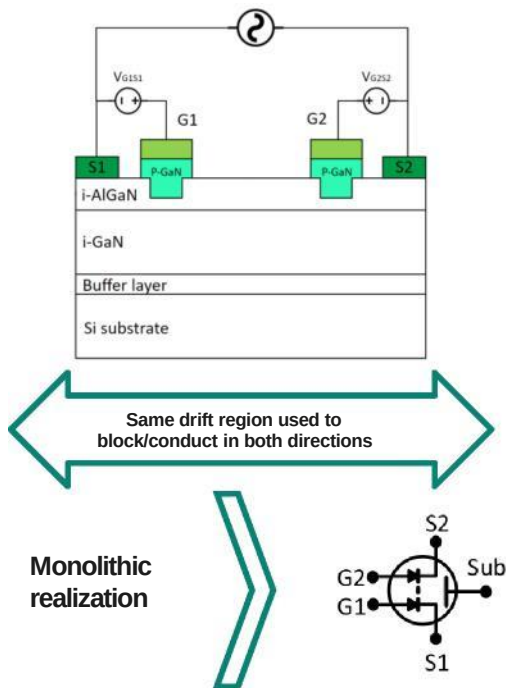
Highlights

- High Performance Subsystem (PPCA) for voltage and current control loops with f_{sw} up to 2MHz
 - Dual M33 with 112 KB SRAM (combined for data and program)
 - 0 WS access to all peripheral registers (ex ADC, Comparator, timer, etc)
 - 4x 25 MSPS ADC, Diff/SE with 4x AFE (max gain 64)
 - Max 34 analog inputs possible
 - 9x Comparator with 12-bit DAC and slope generator functionality
 - 2x 12-bit DAC with buffer for analog output
 - HW filters like median, average, moving average for ADC post processing
 - HW filters for computing 3p3z functions
 - HW block for current observer and matrix commutation
 - Dynamic and low latency trigger and event handling through EPU
 - 0 waitstates cycle CORDIC accelerator for trigonometric functions
 - < 100ps high resolution PWM (HRPWM)
- Memory:
 - 128-bit RWW Flash @ 20MHz, ECC on Flash & SRAM (128KB)
 - 16 kB Instruction Cache
- Main CPU, M33 for housekeeping and PMBUS handling
- Serial communication with PMBUS compliant and support up to 1Mbps
- Security: PSA L3 certified, Configurable Flash Partitioning/Protection
- Safety: Class B & SIL 2 compliant

Operating range

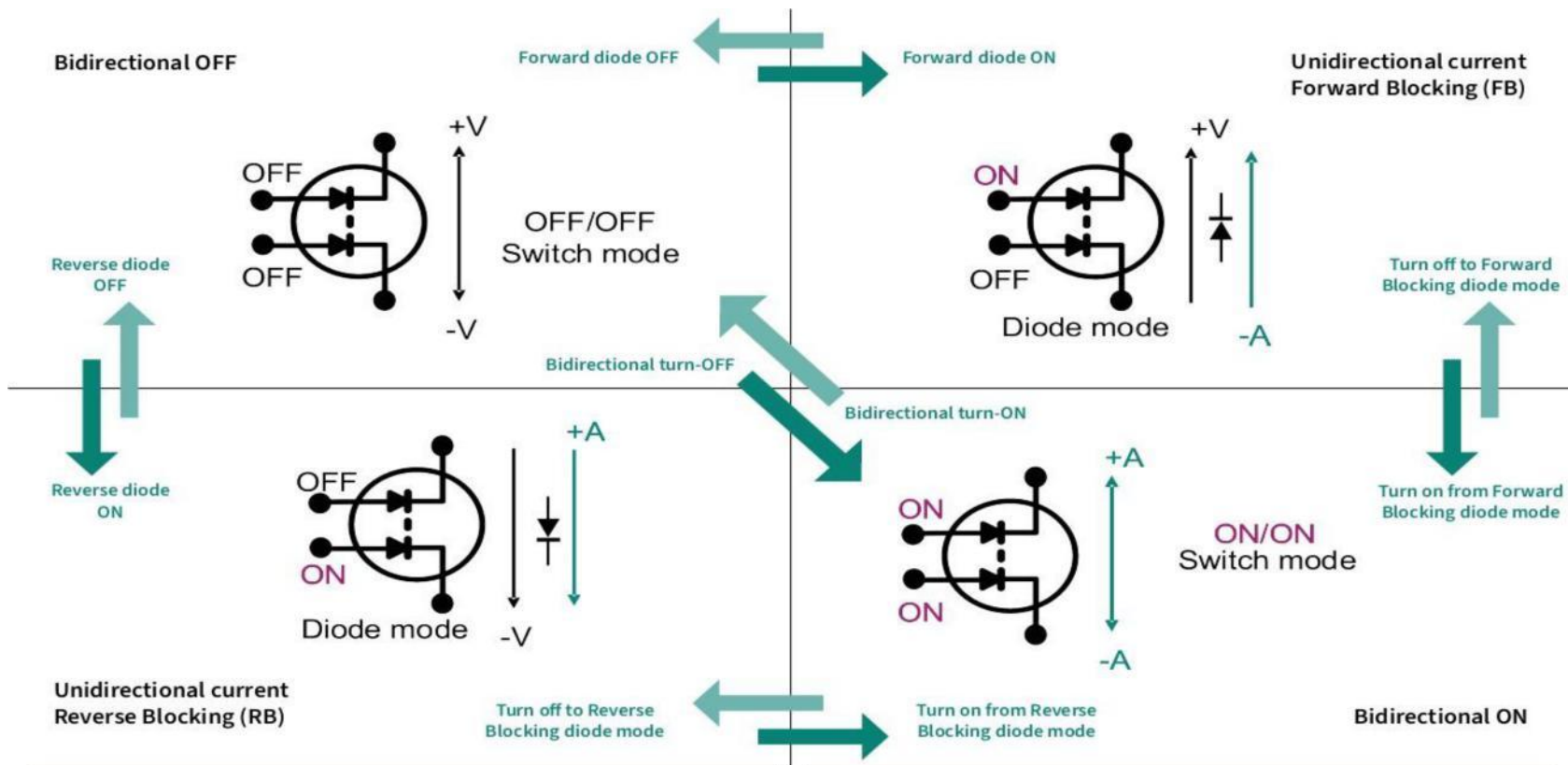
- 2.6 – 3.3V
- <1 – 10 μ A low power modes
- 40 – 105°C T_a

Infineon's CoolGaN high voltage bidirectional switch (BDS)



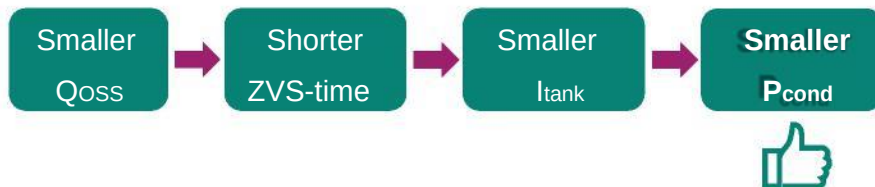
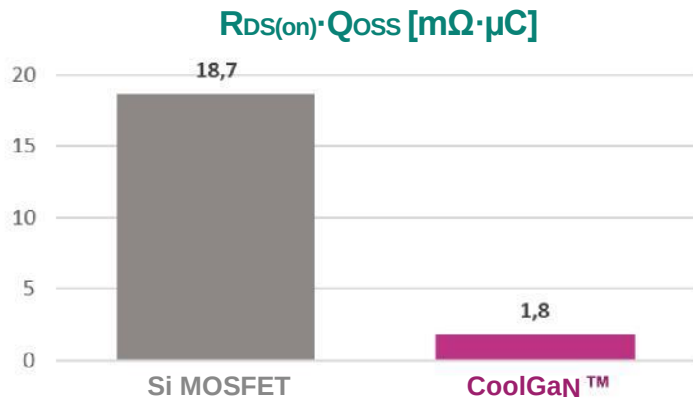
- The **CoolGaN high voltage bidirectional switch (BDS)** is a monolithic GaN switch that can block voltage and current in both directions, featuring a double gate structure and leveraging Infineon's rugged Gate Injection Transistor (GIT) technology.
- Its unique architecture enables the use of the same drift region to block voltages in **both directions**, resulting in a reduced die size compared to conventional back-to-back arrangements.
- The **CoolGaN BDS** boasts four active terminals and an additional substrate terminal, offering **four modes of operation**, including two traditional on/off modes and two diode modes, providing unprecedented control and flexibility for power system design.

Modes of operation for CoolGaN BDS

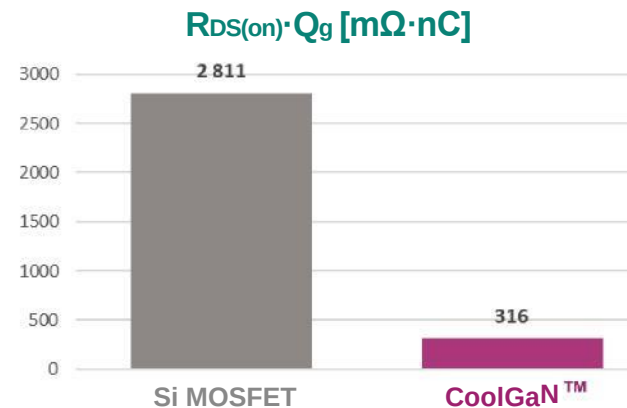


CoolGaN™ superior figures of merit (FOM)

- CoolGaN™ Q_{oss} is substantially smaller than in Si



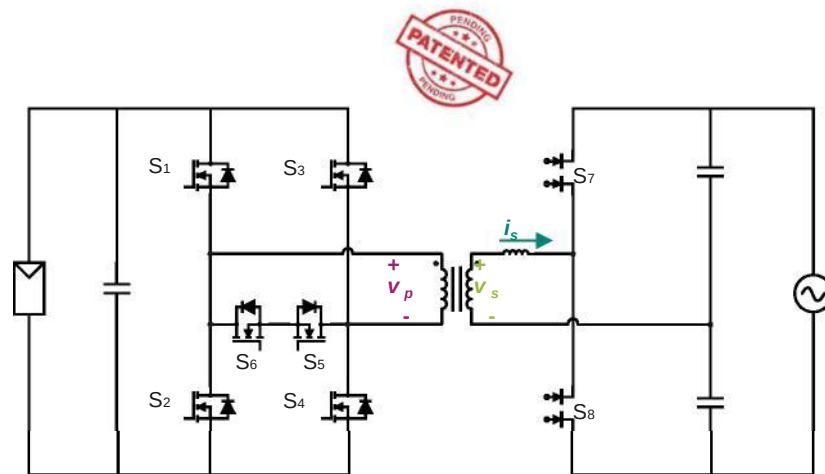
- CoolGaN™ Q_G and Q_{GD} is substantially smaller than in Si and SiC



- Smaller **turn-off losses**
- Enables **soft switching & high frequency**

Infineon cycloconverter topology: basics

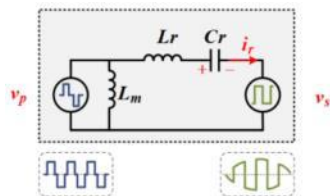
- Cycloconverter topology enables **superior efficiency** thanks to the possibility to achieve soft-switching:
 - ZVS for the power switches at the primary-side
 - and ZCS for the power switches at the secondary
- Furthermore, thanks to the bi-directional switches of the secondary side, **reactive power regulation** can be easily implemented
- By enabling soft-switching, **power density** can be achieved by operating the devices at higher switching frequency (which in turn helps to reduce passive components size and cost)
- Back-to-back switches at the AC-side can be conveniently replaced by **bi-directional GaN switches**



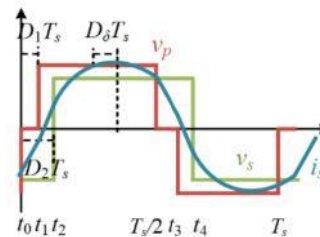
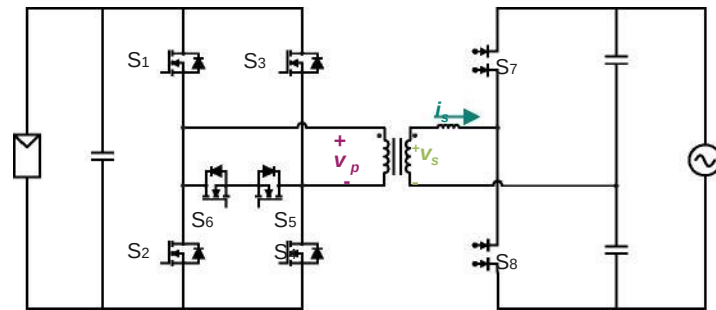
Infiniteon cycloconverter topology: operation 1/2

- The topology features a classic **full-bridge** modulating the energy injected into a **resonant LC tank** through a transformer

- **Tank equivalent** circuit:



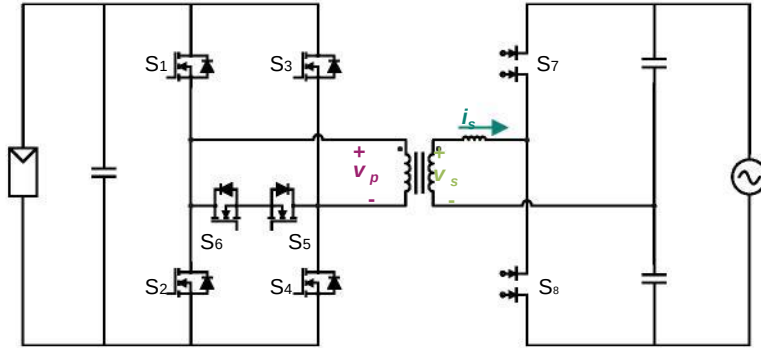
- **Primary side** is modulated through a typical **unipolar modulation** scheme: $v_p = \{ +V_{in}, 0, -V_{in} \}$
- **Secondary side** current bounces between the high-side and low-side bi-directional legs $v_s = \{ +|V_g|/2, -|V_g|/2 \}$
- At different switching frequencies, the **impedance of the LC resonant tank** changes



- Most efficient energy conversion can be achieved by contemporary control of:
 - Primary duty-cycle D_1
 - Phase-shift: D_δ
 - **Switching frequency**

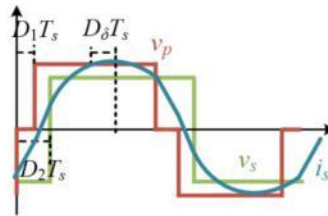
Infinion cycloconverter topology: operation 2/2

- Infineon proprietary (patent pending) Cycloconverter solution for microinverter systems

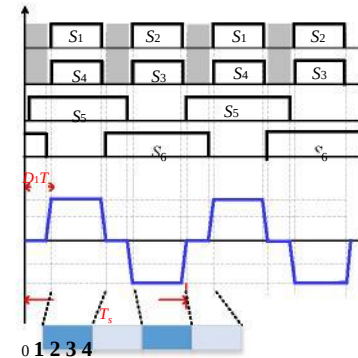


- Modulation parameters:

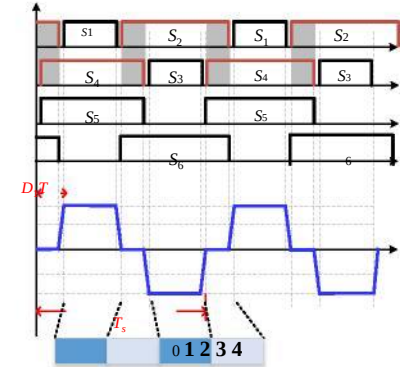
- Primary duty cycle
- Phase-shift primary to secondary
- Frequency



Modulation Scheme 1



Modulation Scheme 2



- Modulation Scheme 2 returns smaller conduction losses per device during the freewheeling-phase

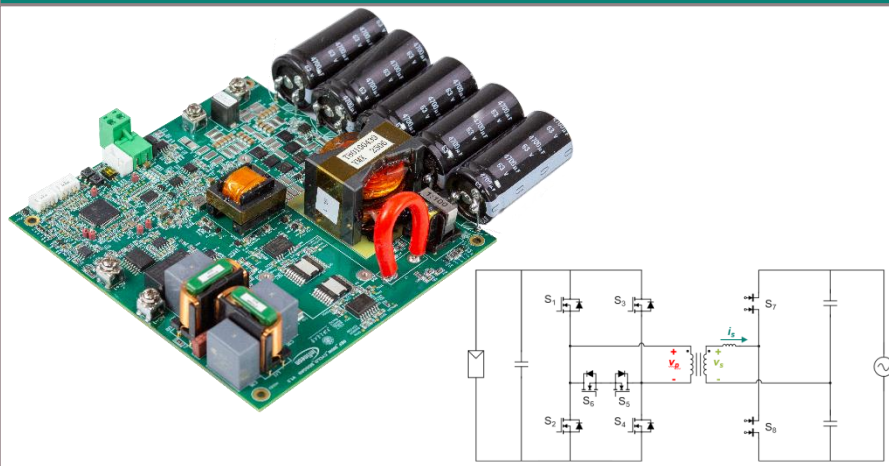
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Microinverter Reference Design Overview

500W innovative reference design using BDS GaN and PSOC™ Control C3 MCU

REF_500W_CYCLO_BDSGAN



Features

- Infineon proprietary Cost effective single stage cycloconverter topology
- CoolMOS™ and CoolGaN™ in soft switching application
- High frequency and power density platform
- **On-grid** operation:
 - **Burst-mode** at light-load
 - **MPPT**
- **Off-grid** operation
- **Protection:** over-voltage, over-current and over-temperature
- Safe **disconnect relay**

Technical details

Target application	Single PV Panel Microinverter
Input voltage range	25 V – 50 V
Output voltage range	240 V L-L, 50/60 Hz
Output current max	2.1 A
Output power max	500 VA
Switching frequency	from 100 KHz to 350 KHz
Board dimension	190 x 145 x 35 (mm)

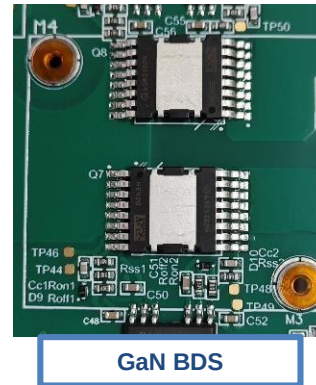
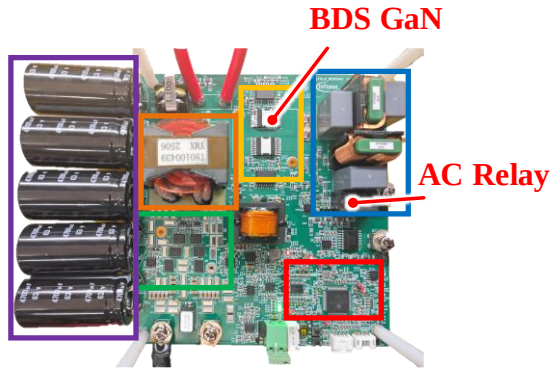
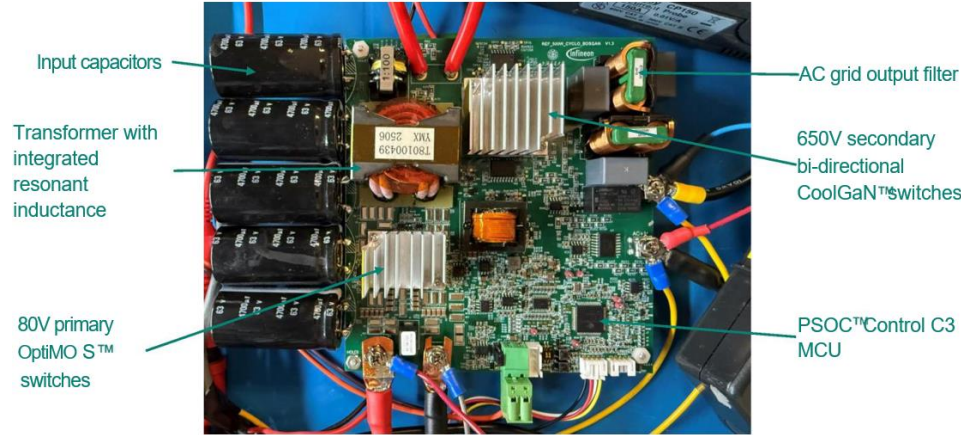
Learn more

Infineon components	– CoolGaN™ BDS: IGLT65R055B2 x2
	– OptiMOS™ 6: ISC014N08NM6 x4 + ISC056N08NM6 x2
	– EiceDRIVER™: 2EDL8034F5B x2, 1EDN8550B x2 - 2EDR7259X x2
	– PSOC™ Control C3 - Main Line: PSC3M5FDS2AFQ1
Sales name	REF_500W_CYCLO_BDSGAN
Release date	Coming soon

Benefits

- **Eff_{peak} ~ 97.6%**
- **EffCEC = 96.7%**
- **iTHD < 5% @half load - <3% @full load**
- Reactive power control: **PF = -0.85 to +0.85**
- **Burst-mode** improves light-load efficiency

Infineon cycloconverter design for microinverter Board demonstrator



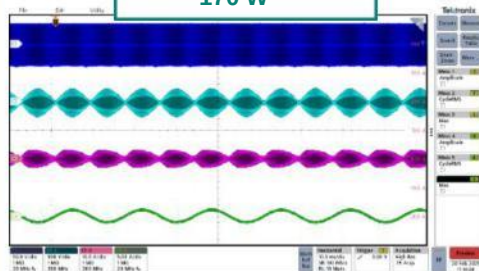
Experimental results

Grid-tied and Off-grid modes

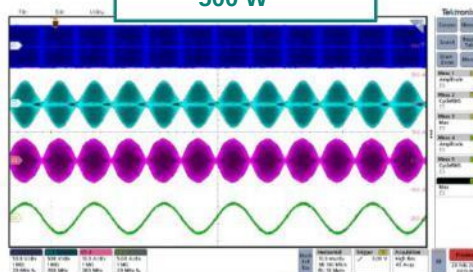
Grid-tied:

- $V_{DC} = 35\text{ V}$
- $PF = 1$

170 W



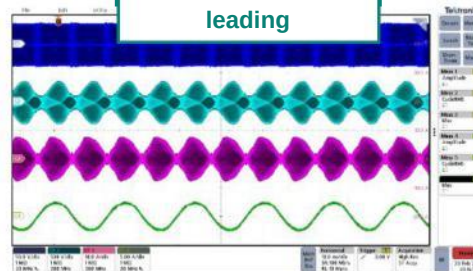
500 W



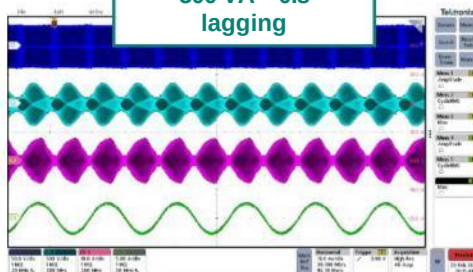
Grid-tied:

- $V_{DC} = 35\text{ V}$
- $PF = 0.8$

500 VA – 0.8
leading



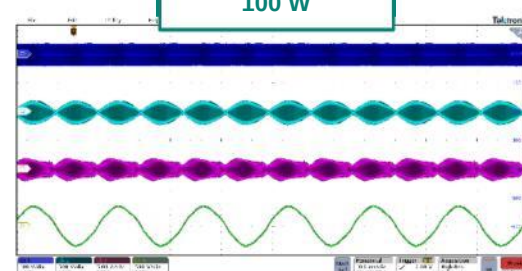
500 VA – 0.8
lagging



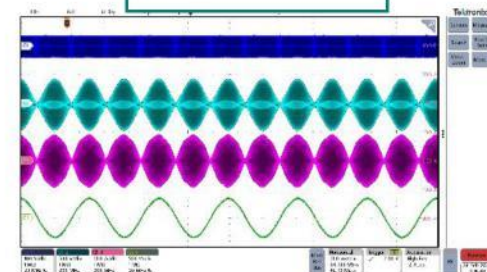
Off-line/R-load:

- $V_{DC} = 35\text{ V}$
- $V_{AC} = 240\text{ V}_{AC}$

100 W



500 W



Waveforms details example – Burst-mode

– $V_{IN} = 35\text{ V}$ – $V_{OUT} = 240\text{ V}_{AC}$

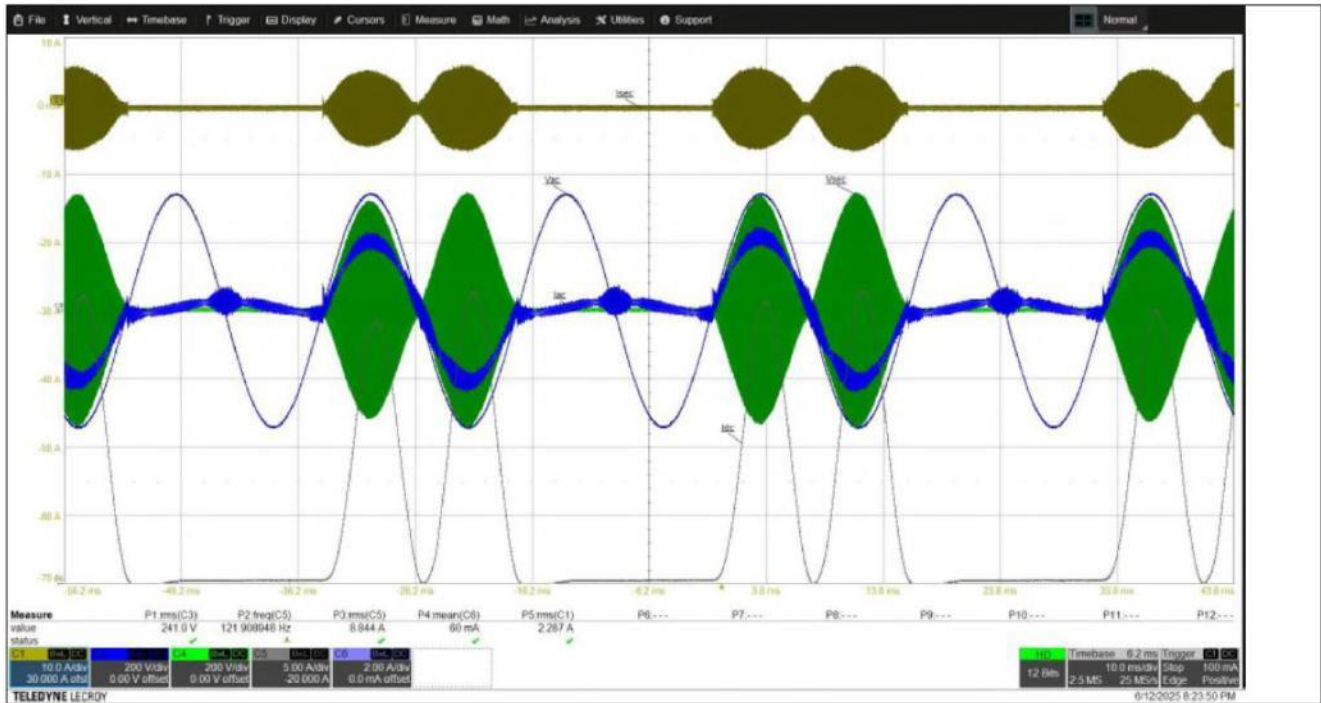
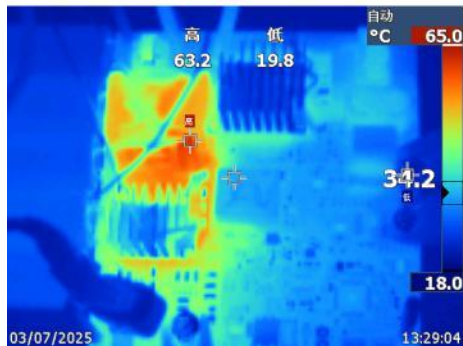


Figure 145 Load 0.75 ARMS, PF 1.0 - grid voltage (blue), output current (purple), DC input current (gray), secondary voltage (green), secondary current (brown)

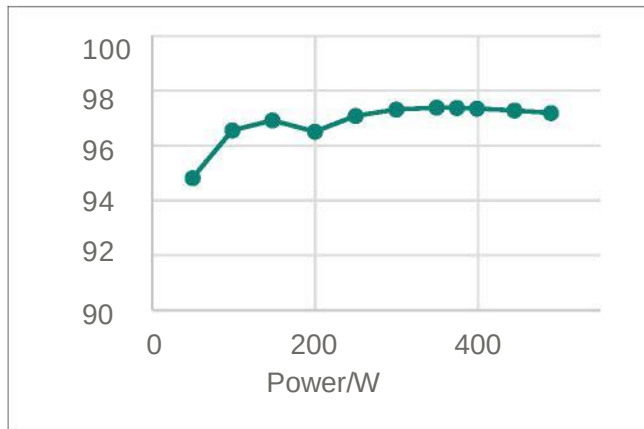
Experimental efficiency and THD

Room temp~18°C, V_{in} =35 V, V_{out} =240 VAC, 30min at 500 W, natural convection, auxiliary power supply powered by PV panels



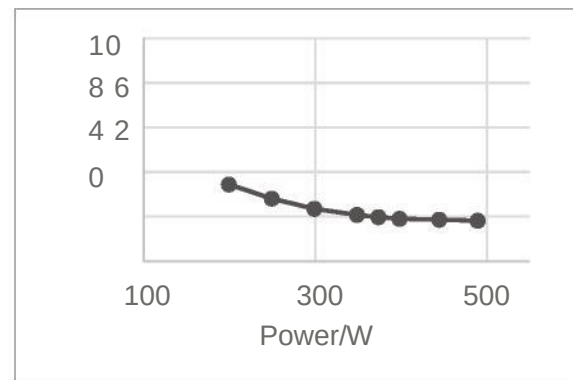
Power analyzer: PW8001

Efficiency (incl. aux. supply)



Peak efficiency: 97.37%
CEC efficiency: 97.1%

THD of output current



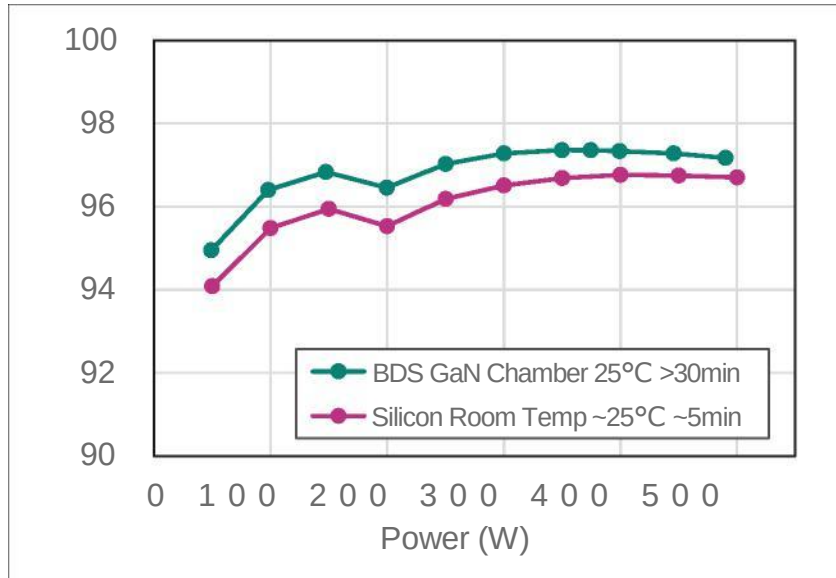
THDi<3% for >50% load condition

Experimental efficiency and THD

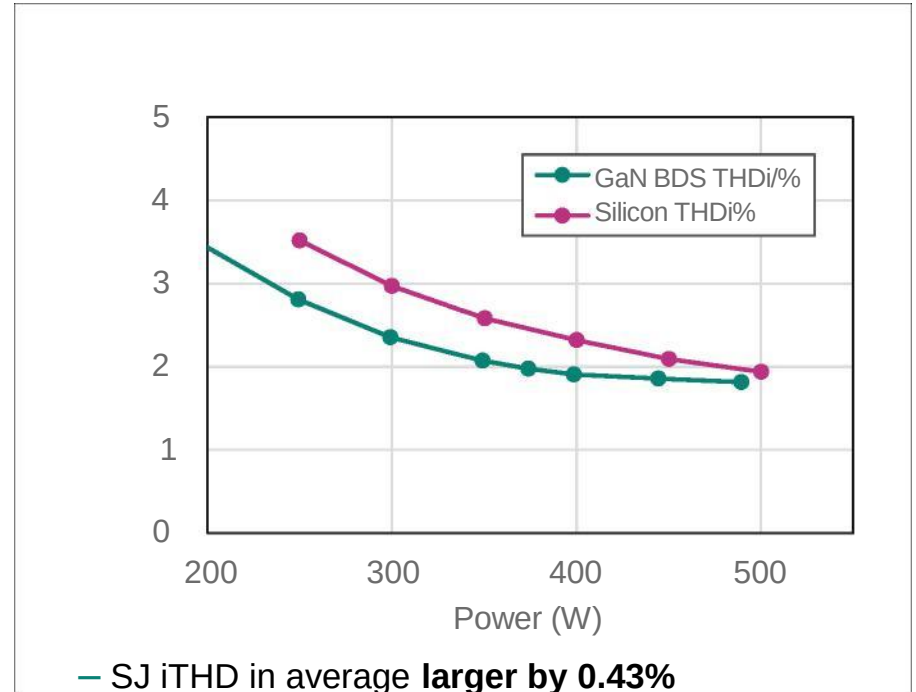
GaN vs CoolMOS

Comparison GaN BDS vs Si-SJ:

- 2x CoolGaN BDS – 650 V – 55 mc) (IGLT65R055B2)
- 4x CoolMOS G7 – 600 V – 43 mc) (IPT60R050G7)



- SJ efficiency in average **lower by 0.87%**
- SJ requires a **larger turn-off current** compared to GaN due to its larger Coss



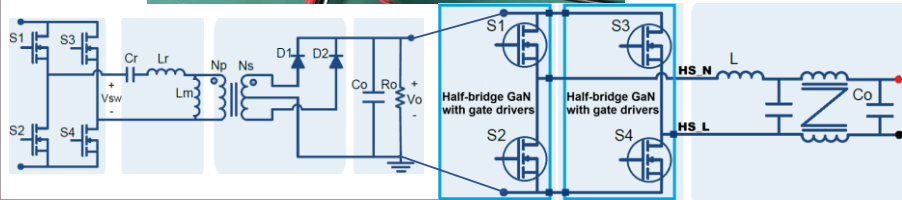
- SJ iTHD in average **larger by 0.43%**
- SJ requires a **larger deadtime** (400 ns) compared to GaN (100 ns) causing distortion on the AC

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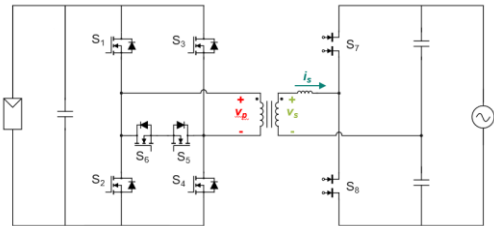
Efficiency comparison

300W_LLC+Inverter



Vin	Inverter Pout	Efficiency
30V	300W (103Vac2.92A)	93.42%
33V	300W (111Vac2.69A)	94.22%
36V	300W (124Vac2.42A)	94.72%
40V	300W (140Vac2.14A)	95.11%
42V	300W (143Vac2.09A)	94.42%

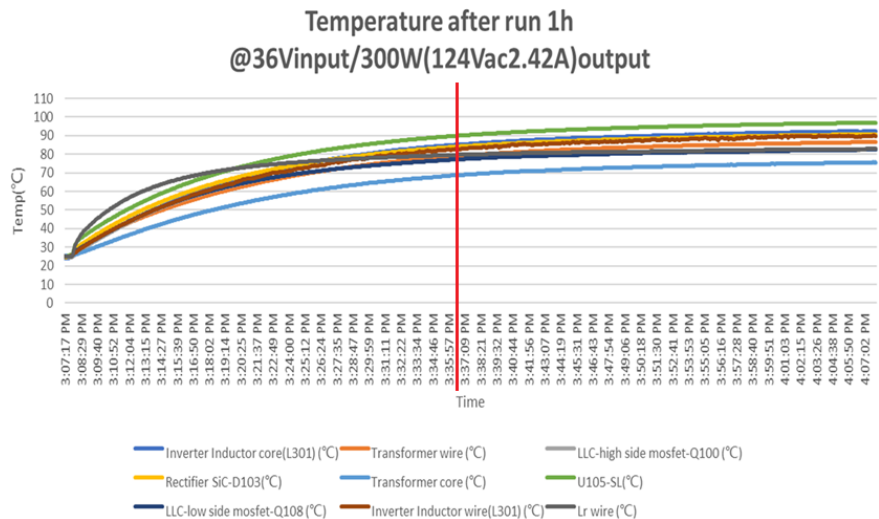
REF_500W_CYCLO_BDSGAN



- Input ~ 35V
- Output ~ 240Vac 500W
- $\text{Eff}_{\text{peak}} \sim 97.6\%$
- $\text{Eff}_{\text{CEC}} = 96.7\%$
- $i_{\text{THD}} < 5\%$ @half load - $< 3\%$ @full load
- Reactive power control: $\text{PF} = -0.85$ to $+0.85$
- **Burst-mode** improves light-load efficiency

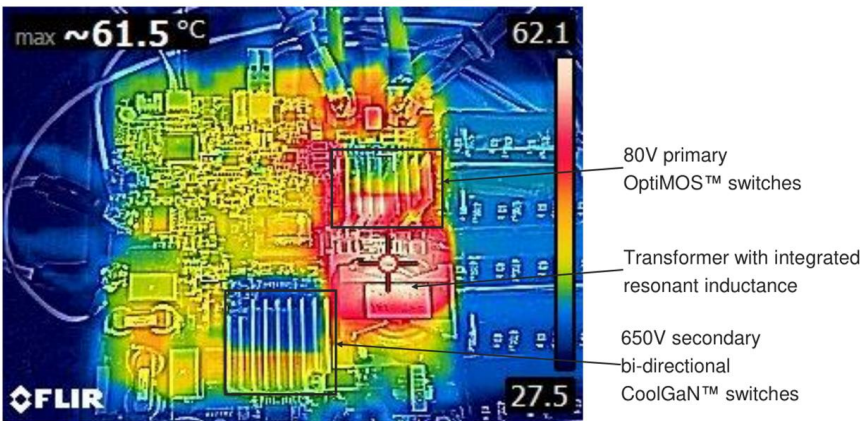
Thermal comparison

300W_LLC+Inverter(run 30mins)



Transformer core(Tmax=70°C)
LLC high side mosfet(Tmax=80°C)
LLC low side mosfet (Tmax=80°C)
Inverter fast leg GaN(Tmax=90°C)

REF_500W_CYCLO_BDSGAN inverter (run 30mins)



Transformer core(Tmax=61.5°C)
Primary side mosfet(Tmax=60°C)
BDS GaN(Tmax=40°C)

Key components comparison

		300W_LLC+Inverter	500W_CYCLO Inverter
Primary side	LLC controller	ICE2HS01G *1	
	Gate driver	2EDL8034G4BXTMA1 *2	2EDL8034F5BXUMA1 *2
	Mosfet	BSC039N06NS *4	ISC014N08NM6 *4
	Transformer	PQ3225 +PQ2016	PQ4035
Seconday side	Rectifier	IDL06G65C5 *2	
	Gate driver		2EDR7259X *2
	MCU	1	Psoc *1
	HV GaN	HB IGI60L1111B1M *2	BDS IGLT65R055B2 *2

